

Why parliamentary systems collapse under fragmentation: A four-institution agent-based model of legislative passage

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Abstract

Parliamentary systems pass more bills than presidential systems at baseline, but collapse to near-zero passage under party-system fragmentation. The literature offers three competing micro-explanations: coalition-formation failure, party discipline, and committee gatekeeping. These mechanisms operate simultaneously in any real legislature, so observational studies struggle to separate their contributions. We present an agent-based model that compares four democratic institutions: pure parliamentary, pure republican/presidential, premier-presidential (France), and president-parliamentary (Russia). The two semi-presidential variants are reachable from a single model class through configurable toggles on government formation and presidential dismissal power. Across four scenarios (baseline, fragmented, polarised, small-system) and $N = 200$ seeds per cell we report bootstrap confidence intervals, Morris screening, Sobol variance decomposition, and mechanism ablations. Two findings emerge. First, party discipline is the mechanism majority-dependent institutions use to aggregate votes; disabling it restores parliamentary passage from 0.05% to 46.7% under fragmentation, and the rescue magnitude is monotone across the four institutions in a pattern that survives varying the common discipline level. Second, the passage–representation tradeoff is a single spectrum: parliamentary maximises legislative throughput at the cost of representational fidelity; republican maximises fidelity via the presidential veto at the cost of throughput; semi-presidential variants split the difference. Together these reframe the fragmentation-collapse literature: the collapse is a discipline problem amplified by majority-rule government formation, not a coalition-formation pathology on its own.

Keywords: legislative studies, institutional design, party discipline, coalition formation, semi-presidentialism, agent-based modelling, Morris screening, Sobol sensitivity.

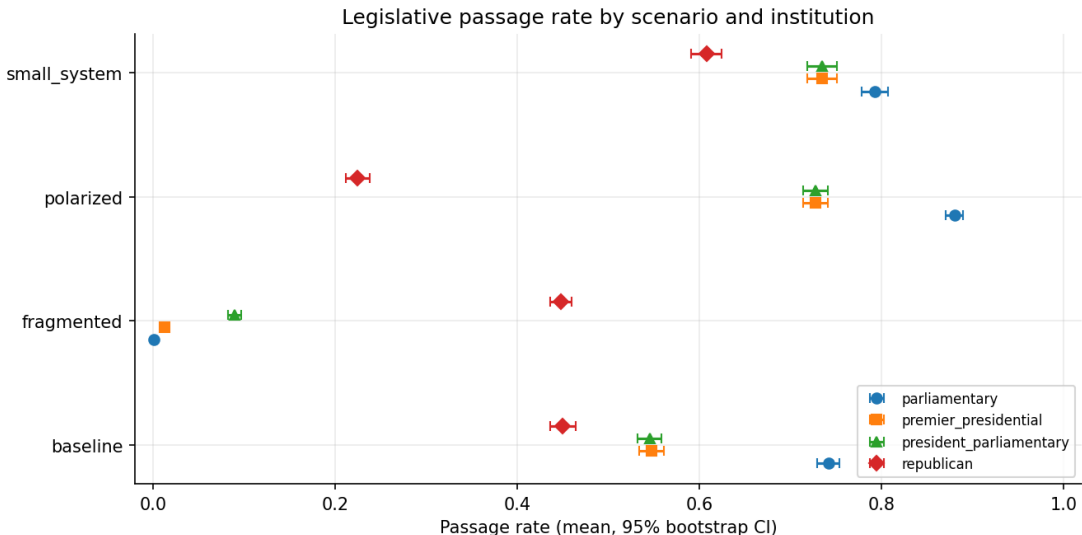


Figure 1: Mean passage rate with 95% bootstrap confidence intervals across four institutions and four scenarios ($N = 200$ seeds per cell). The baseline parliamentary advantage reverses under fragmentation: pure parliamentary and premier-presidential collapse to near-zero passage, while president-parliamentary partially rescues and republican is unaffected. Section 4.1 reports the full passage-rate table; Section 4.4 attributes the collapse to party discipline.

1 Introduction

Democracies differ dramatically in how much legislation they produce. In the United Kingdom, over 95% of government bills reach Royal Assent and government defeats on second reading have not occurred since 1986 [10]. Germany passes roughly 90% of government-initiated bills [3]. In contrast, only 3–7% of bills introduced to the U.S. Congress are enacted into law, though most of that figure reflects filtering at committee stage rather than genuine floor rejection [7]. France, a premier-presidential system, enacts roughly 70% of its laws through government-sponsored *projets de loi* [6]. Russia’s State Duma, a president-parliamentary system, passes essentially 100% of presidential bills while opposition initiatives succeed at below 1% [13].

These differences reflect institutional rules rather than underlying preferences: the cross-national variation in legislative productivity is much larger than the cross-national variation in what electorates want. But the rules interact in ways that are difficult to disentangle. Three mechanistic accounts compete to explain why some legislatures produce laws efficiently and others grind to a halt. The first emphasises **coalition formation**: where a single party cannot hold a working majority, bargaining over portfolios and policy may fail to produce a viable government, blocking any legislative action before it starts [11, 18]. The second emphasises **party discipline**: strong whipping aligns MPs with their leadership rather than their personal or constituent preferences, enabling the government to pass bills that a free vote would reject [1, 2]. The third emphasises **committee gatekeeping**: structural rules that allow small numbers of legislators to kill bills before floor consideration [15, 19]. These accounts are not mutually exclusive, but their relative contributions to any observed passage pattern are difficult to estimate because all three operate simultaneously in every real legislature.

This paper uses an agent-based model (ABM) to attribute passage outcomes to specific mechanisms. An ABM can implement coalition formation, party discipline, and committee gatekeeping in a single deterministic simulation and disable them one at a time while holding everything else fixed. The mechanism-level counterfactuals required for causal attribution are straightforward in a simulation and impossible in observational data. Prior ABM work in legislative studies has typically modelled a single mechanism at a time: coalition formation [11], electoral mandate structure [5], and spatial voting in the Downs [4] tradition. Our contribution is to integrate the three competing mechanisms in one framework and compare their contributions across institutions and scenarios, building on the stylised pattern that presidential systems achieve lower legislative success rates than prime-ministerial systems while remaining robust to coalition failure [14, 16].

We compare four institutional types. Pure **parliamentary** systems follow the Westminster-Continental European template: the cabinet is drawn from the legislature and falls on failed confidence votes. Pure **republican** (presidential) systems follow the U.S. template: the executive is elected separately, has formal veto power, and is not responsible to the legislature. Between these two sit the **semi-presidential** variants classified by Shugart and Carey [17]. **Premier-presidential** systems (France, Portugal, Ireland) have a directly elected president alongside a cabinet responsible to parliament, but the cabinet is formed by parliamentary majority and the president cannot unilaterally dismiss the prime minister. **President-parliamentary** systems (Russia, Weimar Germany) have the same pair of offices but the president forms the government and can dismiss the PM without a parliamentary vote. In our model both variants are reachable from

a single class through two independent configuration toggles.

Across these four institutions we run $N = 200$ random seeds per scenario, report bootstrap 95% confidence intervals, use Morris elementary-effects screening and Sobol variance decomposition to identify the dominant parameters, and run mechanism ablations (disabling committees, discipline, or veto one at a time) to attribute passage contributions. We also check the robustness of the main ordering by varying the common discipline level applied to all institutions.

Two findings structure our contribution. First (§4.4), under fragmentation the *no-discipline* ablation restores parliamentary passage from 0.05% to 46.7%, and the rescue magnitude is monotone across the four institutions in the order parliamentary > premier-presidential > president-parliamentary > republican. Under polarisation the same monotone ordering holds in magnitude but with the opposite sign: disabling discipline now costs parliamentary 66 percentage points of passage. The sign-flip reveals the underlying mechanism: discipline is the tool majority-dependent institutions use to aggregate votes, enabling passage when a cohesive coalition can enforce its will and blocking passage when it cannot. Second (§4.5), the passage–representation tradeoff is a single spectrum. Parliamentary passes the most bills but its passed-bill ideology matches no particular filter (policy representation gap $\approx +0.04$ at polarised). Republican filters aggressively through the presidential veto (gap ≈ -0.53 at polarised) at the cost of high non-passage rates. Semi-presidential variants sit between.

The remainder of the paper proceeds as follows. Section 2 describes the model; Section 3 describes the experimental design; Section 4 presents the results; Section 5 discusses implications and relates the findings to prior work; Section 6 states limitations; Section 7 documents code, data, and the interactive companion app.

2 Model description

2.1 Agents and state

The model contains four agent types: **constituencies** with a fixed two-dimensional ideology (economic $\times$ social) and a population; **parties** with an ideology and a name; **legislators** with an ideology, an assigned constituency, and an assigned party; and **committees** with a policy-area jurisdiction (an ideology centre and radius) and party-proportional membership. Bills are ephemeral objects with an ideology and a salience, instantiated per legislative step.

All four institutions share the same agent types and initialisation procedure. They differ only in the bill-processing pipeline invoked by `pass_legislation(bill)` and in the institution-specific state that pipeline reads and writes (government, cabinet, president, confidence votes, vetoes, dismissals).

2.2 Passage pipelines

Parliamentary: bills route to a committee with matching jurisdiction (committees can approve, amend, or kill); with probability `confidence_matter_rate` a bill is treated as a confidence vote and whipped harder; a floor vote with party discipline decides passage; failed confidence votes trigger government collapse and reformation.

Republican: bills route to a committee (with stronger gatekeeping power); a floor vote with weak discipline decides legislative passage; the executive vetoes probabilistically by ideological distance from the bill; a two-thirds override attempt follows vetoes.

Semi-presidential: hybridises both pipelines. Committees route bills; bills may be treated as confidence matters; a floor vote with moderate discipline decides legislative passage; the directly-

elected president then checks a veto; if the president can dismiss the PM (president-parliamentary variant) the dismissal check fires on every bill and may collapse the government independently of confidence votes.

2.3 Configuration toggles for semi-presidential variants

Two `SemiPresidentialConfig` toggles distinguish the variants:

- `government_formation` $\in \{\text{majority_driven}, \text{president_driven}\}$. Majority-driven formation follows the parliamentary rule (largest party alone if it holds a majority, else two-party coalition). President-driven formation appoints the president’s party as the governing party, adding a second party for majority only if available; otherwise a minority cabinet.
- `president_can_dismiss_pm` $\in \{\text{False}, \text{True}\}$. When `True`, the president dismisses the PM with probability `presidential_dismissal_rate` whenever their ideological distance exceeds 0.5.

Premier-presidential corresponds to `(majority_driven, False)`; president-parliamentary to `(president_driven, True)`. Non-canonical combinations are reachable but not the focus here.

2.4 ODD protocol

A full ODD description [8, 9] appears in Appendix A. All agent states, scheduling, design concepts, initialisation, and submodel equations are documented there.

3 Experimental design

3.1 Scenarios

Four scenarios vary the political environment while holding institutional rules fixed: **baseline** (20 legislators, 3 parties, 6 constituencies, 25 bills, bill ideology range $[-1, 1]$); **fragmented** (24 legislators, 5 parties, 8 constituencies, 30 bills, range $[-1, 1]$); **polarised** (16 legislators, 2 parties, 4 constituencies, 20 bills, range $[-1.5, 1.5]$); and **small-system** (10 legislators, 2 parties, 3 constituencies, 15 bills, range $[-0.8, 0.8]$).

3.2 Statistical harness

Each (scenario, institution) cell is replicated over $N = 200$ seeds. We report means with percentile bootstrap 95% confidence intervals (`scipy.stats.bootstrap`, 9999 resamples). Pairwise institution comparisons use Welch’s t -test and Mann-Whitney U ; effect sizes are reported as Cohen’s d . When zero-variance cells produce undefined p -values, those are recorded as `NaN` rather than imputed.

3.3 Sensitivity analysis

For each institution we define a six-parameter problem and run Morris elementary-effects screening ($r = 20$ trajectories, 4 levels) and Sobol variance decomposition ($N = 256$ base samples, first-order and total-order indices). Three seeds per design point reduce stochastic variance without inflating cost. The six parameters per institution always include `discipline_strength`, `committee_gatekeeping_power`, `num_parties`, and `num_constituencies`; the remaining two are institution-specific.

3.4 Mechanism ablations

Three ablations modify the running model without touching the production code. `no_committees` replaces the committee-routing function with a pass-through; `no_discipline` sets `discipline_strength = 0`; `no_veto` (where applicable) replaces the executive-veto check with `False`. Each ablation is monkey-patched on the live model instance; no class-level mutation occurs.

3.5 Discipline-default robustness check

To test whether the observed ordering depends on our default discipline values, we sweep a common discipline level $D \in \{0.0, 0.1, \dots, 0.9\}$ applied simultaneously to all four institutions, and compute $\text{rescue}(D) = \text{passage}(D = 0) - \text{passage}(D)$ per institution at $N = 100$ seeds per cell.

4 Results

4.1 Passage rates across scenarios

Figure 1 (placed on the title page) summarises per-scenario mean passage rates with 95% bootstrap CIs for all four institutions. At baseline the expected ordering holds: parliamentary 0.742 [0.729, 0.754], premier-presidential 0.547, president-parliamentary 0.546, republican 0.450 [0.436, 0.464]. Parliamentary’s advantage (+29pp) over republican is highly significant (Welch’s $t = 38.9$, $p < 10^{-200}$; Cohen’s $d = 2.88$).

4.2 The fragmentation collapse

Under fragmentation, parliamentary passage collapses to 0.001 (0–0.001 CI) while republican holds at 0.448. The two semi-presidential variants split: premier-presidential collapses to 0.013 (similar to parliamentary because it shares the majority-driven formation rule), while president-parliamentary partially rescues passage to 0.086 through its president-driven minority-cabinet rule. Figure 2 shows the per-seed distributions: the collapse is a regime property, not a tail event.

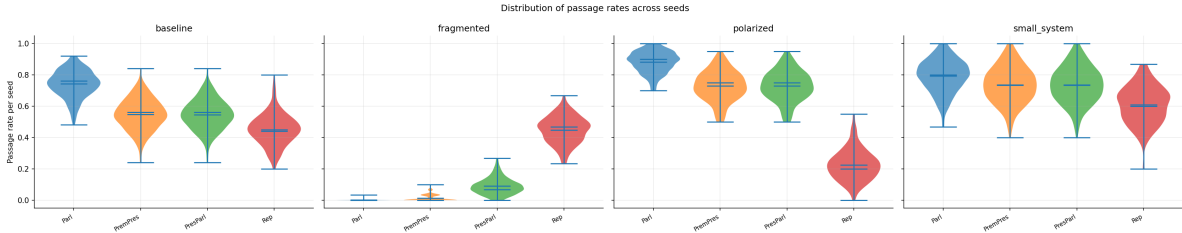


Figure 2: Per-seed passage-rate distributions, faceted by scenario. The fragmented cell shows near-zero passage for both majority-driven institutions (parliamentary, premier-presidential) and a partial rescue under president-parliamentary.

4.3 Sensitivity: different institutions, different bottlenecks

Sobol total-order indices at baseline (Fig. 3) show that parliamentary passage is dominated by `num_parties` ($S_T = 0.808$), consistent with the fragmentation-collapse finding. Republican passage is dominated by `committee_gatekeeping_power` ($S_T = 0.671$) with `discipline_strength` second ($S_T = 0.477$). The two institutions have structurally different bottlenecks. Semi-presidential variants track parliamentary in their dependence on `num_parties`.

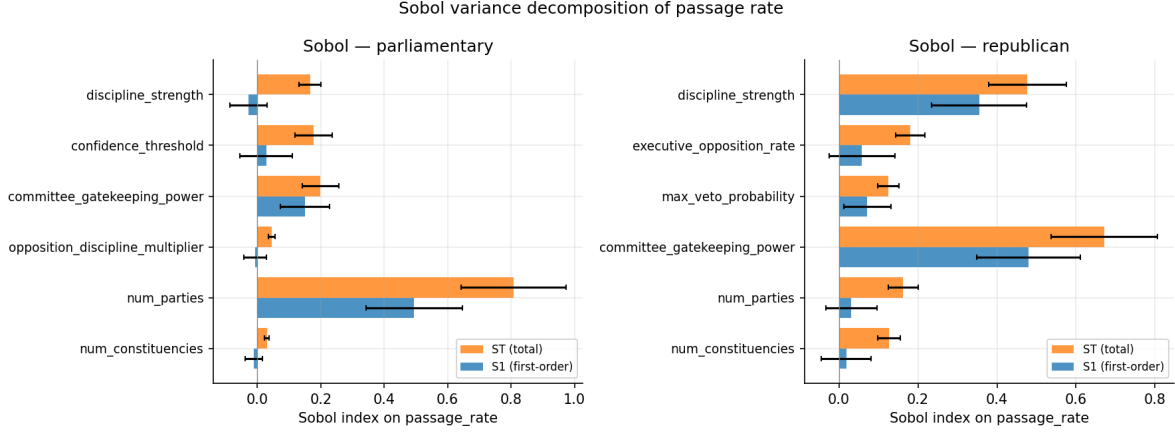


Figure 3: Sobol first-order (S_1) and total-order (S_T) indices for passage rate at baseline.

The same analysis on the fragmented scenario confirms the num-parties dominance across all three majority-dependent institutions ($S_T \approx 0.76$ – 0.79). `confidence_threshold` gains importance under fragmentation (+0.12 in S_T for parliamentary), which is consistent with a binding majority-threshold interpretation. Republican remains structurally insensitive to `num_parties` ($S_T = 0.14$).

4.4 Mechanism attribution via ablation

Figure 4 shows $\Delta \text{passage_rate}$ relative to the full model for each (scenario, institution, ablation) cell at $N = 200$. The headline result appears at fragmented:

Table 1: no_discipline ablation Δ under fragmentation.

Institution	Full-model passage	Δ (no_discipline – full)
parliamentary	0.001	+0.466
premier_presidential	0.013	+0.312
president_parliamentary	0.086	+0.233
republican	0.448	−0.242

Disabling party discipline restores parliamentary passage from near-zero to nearly 47%. The rescue magnitude decreases monotonically across the four institutions in the order parliamentary > premier-presidential > president-parliamentary > republican. Republican is the only institution where zeroing discipline *hurts* rather than helps: without a majority-formation gate, discipline plays a different role (crude vote aggregation on the floor) and removing it simply damages the signal.

Under polarised scenarios the same ordering holds in magnitude but with opposite sign (Table 2). Discipline *enables* passage when the coalition wants extreme bills to pass; disabling discipline costs parliamentary 66 percentage points. Republican, with no coalition to whip, loses only 11 percentage points. Across scenarios, discipline is the mechanism majority-dependent institutions use to *aggregate votes*, rather than a mechanism that specifically causes fragmentation pathologies.

4.5 Passage–representation tradeoff

Figure 5 plots passage rate against a new metric, the policy representation gap:

$$\text{gap} = \overline{d_{\text{passed}}} - \overline{d_{\text{proposed}}} \quad (1)$$

Table 2: no_discipline ablation Δ under polarisation.

Institution	Full-model passage	Δ (no_discipline – full)
parliamentary	0.880	−0.664
premier_presidential	0.728	−0.557
president_parliamentary	0.723	−0.557
republican	0.225	−0.113

where d is the L2 distance between a bill’s ideology and the constituency-median ideology. Negative gaps indicate the institution filters passed bills toward the median; positive gaps indicate the institution spreads passed bills relative to random proposal draws.

The tradeoff is sharpest under polarised scenarios:

Table 3: Passage–representation tradeoff at polarised ($N = 200$).

Institution	Passage rate	Representation gap
parliamentary	0.882	+0.037
premier_presidential	0.725	−0.012
president_parliamentary	0.730	−0.010
republican	0.218	−0.530

Parliamentary passes 88% of bills with essentially no ideological filter; republican passes only 22% but the bills that do pass are strongly biased toward the constituent median, reflecting the presidential veto’s role as an ideological screen. Semi-presidential variants sit on the same spectrum between them. Filtering and throughput are mechanistically linked: any mechanism that increases one reduces the other.

4.6 Robustness of the monotone ordering

The ablation ordering in Table 1 could in principle be an artefact of our default `discipline_strength` values, which happen to be ordered the same way across institutions (0.8, 0.6, 0.6, 0.4). Figure 6 tests this by setting all four institutions’ `discipline_strength` to the same value $D \in \{0.1, \dots, 0.9\}$ and measuring the rescue magnitude.

The monotone ordering parliamentary > premier-presidential > president-parliamentary > republican holds at 9/9 tested D values under fragmentation. The structural claim survives: the ordering is not a function of our chosen defaults. A regression test in the repository enforces this at $D \in \{0.0, 0.2, 0.5, 0.8\}$ so that future mechanism changes that break the ordering fail CI.

5 Discussion

5.1 Discipline as a universal aggregation mechanism

The ablation results across fragmented and polarised scenarios (Tables 1 and 2) reveal that party discipline is not simply a “collapse trigger” under fragmentation or a “passage enabler” under polarisation. It is the same mechanism in both cases: discipline aggregates legislators toward the coalition-leadership preference rather than the free-vote median. In regimes where the coalition can form and its preferences are near the floor median, discipline enforces passage. In regimes where the

coalition cannot form, discipline enforces an imaginary position that blocks passage. The sign-flip between the two scenarios is a direct consequence of this mechanism, not evidence of two separate mechanisms.

The monotone ordering across the four institutions reflects how much each relies on a formation-gated coalition to act. Parliamentary requires a majority coalition before any legislation is possible; premier-presidential requires the same, but with a presidential veto adding a second gate; president-parliamentary also nominally requires a majority, but the president-driven formation rule lets the government function as a minority cabinet when no majority forms; republican has no formation gate at all. The rescue magnitude under `no_discipline` scales with this dependence. This reading is consistent with Strøm [18]’s argument that minority governments can function when institutions permit, and with Shugart and Carey [17]’s typology of semi-presidential variants.

This framing also makes a prediction that a pure “fragmentation-collapse” framing does not: under polarisation, the same institutions that suffered the fragmentation collapse should benefit most from discipline. Table 2 confirms this.

5.2 The passage–representation tradeoff as a single spectrum

Figure 5 and Table 3 make a concrete claim about the tradeoff between legislative throughput and representational fidelity: the two quantities lie on a single axis for the four institutions studied. Parliamentary sits at one end (high throughput, low filter); republican at the other (low throughput, high filter); semi-presidential variants between them. The mechanism is straightforward. Discipline-plus-majority passage, the parliamentary mechanism, has no ideological test: whatever the coalition proposes passes. The presidential veto, the republican mechanism, *is* an ideological test: bills far from the executive’s ideology are filtered probabilistically. Committees add filtering in both but the effect is smaller than the presidential veto.

This casts the parliamentary “advantage” in baseline passage rates differently. Parliamentary systems pass more bills in absolute terms but do not necessarily pass *better* bills in the sense of matching constituent preferences. The comparison requires joint consideration of both axes, not rank-ordering on one.

5.3 Policy and institutional-design implications

The conventional response to fragmented-parliament gridlock is *electoral reform*: raise thresholds, move toward first-past-the-post, encourage larger parties. Our results suggest a second route: *discipline reform*. Allowing free votes on more bills, weakening leadership whip powers, or restructuring MP incentives toward their districts rather than their parties could restore passage capacity under fragmentation without touching electoral rules. Whether such reforms also damage representation is an open empirical question: the parliamentary filter is already weak in our model, so weakening discipline further might spread passed bills without the presidential-veto compensation. This model can illuminate that question but cannot yet answer it directly.

For new constitutional design, the Phase F robustness result suggests a trade-off between fragility and filtering. Pure parliamentary maximises passage at baseline but has a failure mode under fragmentation; president-parliamentary partially insulates against that failure mode but sacrifices some passage at baseline; republican is robust at all scenarios but pays a throughput tax. Premier-presidential is closer to parliamentary than its name suggests and shares the fragmentation vulnerability.

5.4 Relation to prior work

Linz [12] argued parliamentary systems outperform presidential systems in producing governments and policy. Our baseline results (0.742 vs. 0.450) echo that finding on passage rates, but our fragmentation results reverse it: parliamentary collapses while presidential holds steady. The conditional reversal recovers a pattern Linz’s categorical claim cannot predict. Shugart and Carey [17] argued semi-presidential variants should differ from each other in specific ways; we confirm that president-parliamentary’s partial rescue of passage under fragmentation is a direct consequence of the president-driven formation rule. Tsebelis [19] argued additional veto players reduce policy mobility; our tradeoff finding refines this by showing that veto players also improve representation quality, so the choice is not “veto players good or bad” but “veto players trade throughput for filtering.”

6 Limitations

The model treats legislator ideology as static and does not include learning or adaptation. It has no endogenous party emergence, realignment, or exit; the party structure is fixed at initialisation. It has no bicameralism, judicial review, federalism, or presidential agenda-setting powers beyond the veto. Bills have no path dependence: each bill is an independent draw. We do not model pre-parliamentary agenda control (e.g., cabinet screening of government bills before introduction), which is why our parliamentary baseline rate of 74% is below the empirical UK and German benchmarks of 90–95% for government bills.

Cohabitation is detected in our semi-presidential runs (shares of 12–26% across scenarios) but does not produce meaningful passage effects: our dismissal rate of 5% per bill under president-parliamentary is too gentle to create sustained gridlock, and the premier-presidential presidential veto is too soft to differentially block bills under cohabitation. Future work should treat cohabitation dynamics more carefully, including cabinet-to-legislator ideological alignment and veto thresholds that shift under divided government.

The sensitivity analysis was run primarily on the baseline scenario; a secondary run on the fragmented scenario (§4.3) confirms the `num_parties` dominance but is not a substitute for per-scenario sweeps of all parameters. The Morris and Sobol samplers include `num_parties` and `num_constituencies` in the sweep space; scenario definitions are therefore overlapping with the sensitivity space, which is a methodological choice to allow continuous interpolation rather than a discrete scenario lookup.

7 Reproducibility

All results are deterministic under seed. The repository <https://github.com/tofuadmiral/institutional-representation-abm> contains the full code, 68 automated tests (Python 3.13 on Ubuntu, GitHub Actions CI), a pinned regression fixture, a Streamlit interactive companion application, and a CoMSES metadata card. The command

```
python -m experiments.multiseed_comparison --scenarios all --seeds 200
```

reproduces Figure 1. Tables 1 and 2 require additional invocations of `experiments.ablation`; Figure 6 requires `experiments.discipline_robustness`; and Figure 5 requires `experiments.representation`. All entry points are documented in the repository `README.md`. Full runtime is under five minutes

on eight cores. An interactive Streamlit companion (`streamlit run streamlit_app/app.py`) exposes every tunable configuration parameter as a slider and reproduces the scenario-comparison, parameter-sweep, and ablation workflows used in §4.

A ODD Protocol

The full ODD protocol [8, 9] is available in the repository at `docs/ODD_PROTOCOL.md`. Core elements are summarised in §2.

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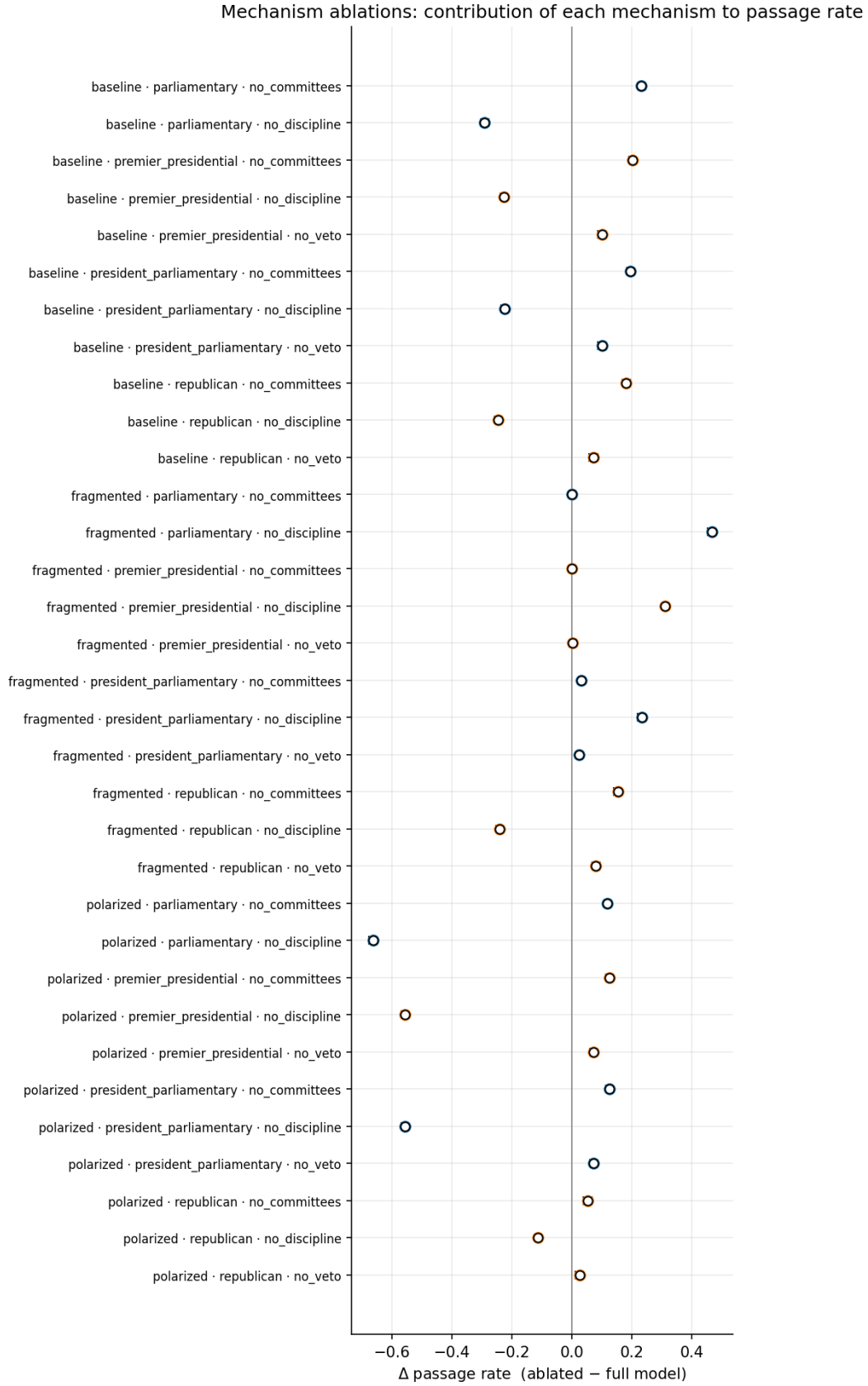


Figure 4: Ablation deltas (Δ passage rate relative to full model) for each (scenario, institution, ablation) cell at $N = 200$ seeds. Confidence bars are ± 1.96 SE.

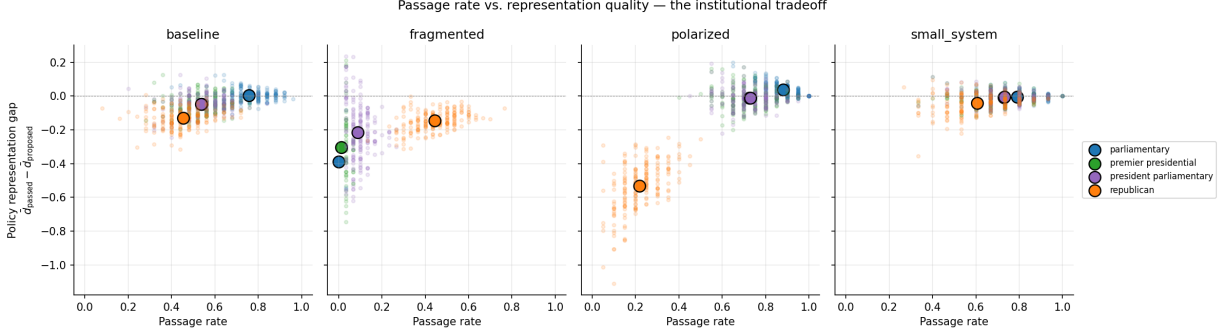


Figure 5: Passage rate versus policy representation gap, per scenario. Each large marker is the $N = 200$ mean; the cloud is per-seed. Under polarisation the tradeoff is most visible: parliamentary at $(0.88, +0.04)$ vs. republican at $(0.22, -0.53)$.

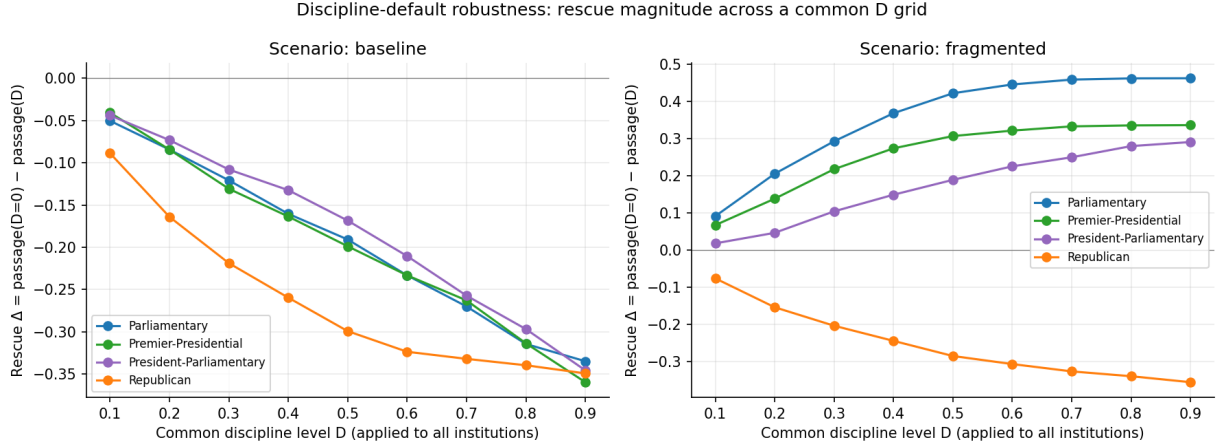


Figure 6: Rescue magnitude versus common discipline level D , per institution, at baseline and fragmented scenarios. Under fragmented (right panel), the monotone ordering parliamentary $>$ premier-presidential $>$ president-parliamentary $>$ republican holds at all nine tested D values.